https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	A is the independent variable and *t* is the dependent variable **or** vary *A* and measure *t*	**1**
	keep $\Delta\theta$ constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • beaker of water • cylinder in water • electrical heater in water • thermometer in water • minimum of three labels from heater, thermometer, cylinder, water, beaker	**1**
	circuit diagram to determine power of the heater e.g. ammeter and voltmeter correctly positioned with a power supply **or** wattmeter correctly connected to power supply and heater	**1**
	method to determine time for temperature of water to increase or *t*, e.g. use a stopwatch/timer	**1**
	method to determine *A*, e.g. micrometer/calipers to determine diameter of cylinder **and** $A = \pi d^2/4$	**1**
	Method of analysis	
	plot a graph of *t* against *A* (**not** logarithmic graphs)	**1**
	$W = \dfrac{\text{gradient} \times P}{h\Delta\theta}$	**1**
	$Z = \dfrac{y\text{-intercept} \times P}{\Delta\theta}$	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 wear (heat proof) gloves to prevent burns from hot beaker/cylinder/heater/water	
	D2 keep *P* and *h* constant	
	D3 check that/ensure/keep initial temperature of the water constant **or** volume/mass of water constant	
	D4 use calipers/ruler to measure *h*	
	D5 repeat measurements of diameter in different directions/at different positions along cylinder and average	
	D6 method to calculate power of heater e.g. $P = VI$ linked to correct circuit diagram for ammeter/voltmeter method	
	D7 repeat measurements of *t* for same *A* and average *t*	
	D8 ensure heater and cylinder are (totally) submerged/immersed **or** stir water (using a glass rod/stirrer)	
	D9 relationship valid if a straight line (not passing through the origin)	
	D10 method to insulate beaker, e.g. use of a lid on the beaker or foam/insulation around outside of beaker	

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